



.22 Winchester Magnum Rimfire

The **.22 Winchester Magnum Rimfire**, also known as the **.22 WMR**, **.22 Magnum**, **.22 WMRF**, **.22 MRF**,^[2] or **.22 Mag**, is a rimfire cartridge. Originally loaded with a bullet weight of 40 grains (2.6 g) delivering velocities in the 2,000 feet per second (610 m/s) range from a rifle barrel, .22 WMR is now loaded with bullet weights ranging from 50 grains (3.2 g) at 1,530 feet per second (470 m/s) to 30 grains (1.9 g) at 2,200 feet per second (670 m/s).^{[3][4]}

History

The .22 WMR was introduced in 1959 by Winchester, but was not used by Winchester until the Winchester Model 61 slide rifle could be chambered for it in 1960.^[5] The first rifle to be offered in the new chambering was the Marlin Levermatic rifle in 1959, because its design was easily modified to accept the more powerful cartridge. By the time of the introduction of the Winchester 61, Smith & Wesson and Ruger had revolvers for it, and Savage had come out with the Model 24 and since late 2012, the Model 42, a more modern update than the 24, a .22/.410 rifle/shotgun combination gun. It was the only successful rimfire cartridge introduced in the 20th century.^[6]

Dimensions and loading

The .22 WMR uses a larger case than the more popular .22 LR, both in diameter and length. The .22 WMR case is a lengthened version of the older .22 WRF and both of these cartridges have the bullet seated within the case, instead of using a heeled bullet as in the .22 LR. In the most common modern .22 WMR loadings

.22 Winchester Magnum



.22 Winchester Magnum Rimfire

Type	Rifle	
Place of origin	<u>United States</u>	
Production history		
Designed	1959	
Manufacturer	<u>Winchester</u>	
Produced	1959–present	
Specifications		
Parent case	<u>.22 Winchester Rimfire</u>	
Case type	Rimmed, straight	
Bullet diameter	.224 in (5.804 mm)	
Land diameter	.219 in (5.6 mm)	
Neck diameter	.2435 in (6.185 mm)	
Base diameter	.2455 in (6.236 mm)	
Rim diameter	.300 in (7.62 mm)	
Rim thickness	.050 in (1.27 mm)	
Case length	1.055 in (26.8 mm)	
Overall length	1.350 in (34.3 mm)	
Rifling twist	1-16"	
Primer type	Rimfire	
Maximum pressure	24,000 psi (170 MPa)	
Ballistic performance		
Bullet mass/type	Velocity	Energy
30 gr. (1.9 g) HP	2,300 ft/s (700 m/s)	322 ft·lbf (437 J)

using a 40-grain (2.6 g) bullet, the combination of more powder and higher sustained pressures gives velocities of 1,875 feet per second (572 m/s) from a rifle and 1,500 feet per second (460 m/s) from a handgun. A .22 WMR round will not fit into the chamber of a .22 LR firearm due to the greater length and width of the case dimensions.

40 gr. (2.6 g) JHP	1,875 ft/s (572 m/s)	324 ft·lbf (439 J)
50 gr. (3.2 g) JHP	1,530 ft/s (470 m/s)	300 ft·lbf (410 J)
Test barrel length: 24 inches (610 mm)		
Source(s): <i>Cartridges of the World</i> ^[1]		

Although the bullet diameters are the same, the larger .22 WMR chamber does not support the smaller .22 LR cartridge. Firing the smaller .22 LR round in a .22 WMR chamber results in swollen or split cartridge cases, high-pressure gas leakage from the rear of the chamber, and bullets striking the chamber throat out of alignment, which can result in injury to the shooter or bystander and which does result in poor ammunition performance.

Gun makers offer .22 revolver models with cylinders chambered either in .22 LR or in .22 WMR and as convertible kits with both cylinders which allow the shooter to easily switch calibers on the same firearm.

Uses

Since the .22 WMR uses bullets comparable in weight to the .22 Long Rifle, but is considerably faster, it shoots flatter and farther and hits harder at all ranges. The same 40-grain (2.6 g) bullet from a .22 WMR at 100 yards (91 m) still has 50% more kinetic energy than .22 LR at the muzzle. The .22 WMR provides improved penetration and more reliable expansion at longer ranges with expanding bullets.^[7]

If sighted in for maximum point blank range on a 3-inch (76 mm) high target, the 40-grain (2.6 g) .22 WMR has an effective range of nearly 125 yards (114 m). This makes the .22 WMR an effective short to medium-range varmint rifle and hunting rifle cartridge. The .22 WMR can effectively kill small game such as rabbits, hares, groundhogs, prairie dogs, foxes, raccoons, opossums, and coyotes. The relatively quiet report and negligible recoil also make it a very pleasant target and plinking round to shoot for extended periods.

Firearms using .22 WMR

It first appeared in the Marlin Model 57M Levermatic carbines in 1959, which was followed by the .22 WMR over .410 bore Savage Model 24 combination gun, followed by Winchester's own Model 61 and 275 pump-action rifles and Model 255 lever-action rifle. The Chiappa Double Badger combination gun also comes with a .22 WMR over .410 bore option. The Springfield Armory M6 Scout is also made in .22 Magnum over .410 bore.^{[8][9]} The Chiappa M6 Survival Gun is similar combination gun that comes with a 12 gauge or 20 gauge shotgun barrel over a .22 Magnum barrel.^[10]

A number of single-shot and repeating rifles were offered in .22 WMR. The .22 WMR operates at pressures beyond what normal blowback actions typically handle, but the self-loading Jefferson Model 159 was introduced for the cartridge.^[5] Until the 1990s, most .22 WMR firearms were bolt-action rifles. In 1977-1985 Harrington & Richardson produced the first American-made semi-automatic .22 WMR. In

the 1990s semi-automatic .22 WMR rifles were also introduced by Ruger (10/22) and Marlin, and are currently produced by Remington (Model 597) and Tanfoglio Appeal Rifle as well as the Excel Arms Accelerator Rifle and Savage arms A22 magnum.^{[11][12][13][14]}

Revolvers in .22 WMR are made by Smith & Wesson, Colt, Taurus, North American Arms, Heritage Arms, and Sturm Ruger.^{[15][16][17][18][19]} Semi-automatic pistols for this cartridge are (or were) produced by Kel-Tec, Grendel and AMT, the latter two now defunct (AMT has been since resurrected by High Standard).^{[20][21][22][23]} The Grendel, AMT and Kel-Tec designs used specially designed chambers with flutes or gas ports, designed to lubricate the long, thin cartridge with gases from the chamber, overcoming the Blish effect and allowing easy extraction of the cartridge.^[24] High-Standard produced various models and versions of their classic two-shot over/under derringer in both .22 WMR and .22 LR.

The Marlin Model 25MG is a smooth bore, eight-shot, magazine-fed, .22 WMR, bolt-action shotgun manufactured by Marlin Firearms.^[25] It was specifically designed to use snake shot, and marketed as a "garden gun" for use in dispatching small garden and farm pests.^[26] It has an effective range of about 15 yards when using snake shot.^[25] Based on earlier Marlin .22 caliber designs, the garden gun was equipped with a high-visibility front sight but no rear sight.^{[27] [28]}

The Argentine EDDA submachine gun uses the .22 WMR round.

In 2014 Kel-Tec released the CMR-30, a PDW-style rifle chambered in .22 WMR with a 16" barrel and a 1:14 twist rate. It uses the same double stack 30-round magazine as the PMR-30.

In 2018 Standard Manufacturing introduced an 8-round .22 WMR revolver at the Las Vegas Shot Show. The revolver was originally introduced as the Volleyfire. The gun has side-by-side barrels, and fires 2 rounds of .22 WMR (1 bullet per barrel) with every pull of the trigger. In 2019 the gun went into production and was renamed "Thunderstruck".

In recent years, a few UK companies such as Guncraft Ltd have been making semi-automatic AR-15-style rifles chambered in 22 WMR, marketed to civilian shooters who, due to the UK's restrictive gun laws, do not have access to self-loading center-fire rifles.^[29]



.22 WMR
snake shot

Ammunition

The .22 WMR is an enlarged, more powerful version of the much earlier .22 WRF.^[5] Despite frequent claims to the contrary, it cannot be safely used in any firearm except those specifically chambered for it. Even firearms chambered for the .22 WRF are not suitable; for one thing, the case lengths are different, and the fact that the cartridge fits into the chamber does not guarantee that using the wrong cartridge is either safe or effective.^[30]

The .22 WMR was for a time the most powerful rimfire round available;^[30] it even outperformed the .22 WCF.^[30] It has since been eclipsed in velocity and overall kinetic energy by the .17 Winchester Super Magnum.

Commonly available in retail stores, the selection and availability of the .22 WMR nevertheless does not match the popular .22 Long Rifle. Furthermore, .22 WMR is typically much more expensive than .22 LR per round, though it is comparable to .17 HMR, which is much more similar in overall performance, and is less expensive than .22 caliber centerfire ammunition.

Because many of the rifles that are chambered for the .22 WMR have tubular magazines, the bullet noses generally are flat or at least blunt to allow smooth feeding and reduce damage to the bullet.^[31] Although a pointed bullet in a rimfire cartridge will not contact the primer of the round in front of it (which is a hazard with centerfire cartridges in a tubular magazine), the manufacturer's stamp is in the middle of the base of a rimfire cartridge, and this may interfere with pointed metal bullets in a tube. However, Remington, CCI, and Hornady now produce bullet designs with 30 or 33-grain (2.1 g) polymer plastic ballistic tips that reduce the hazards of pointed ammunition in tubular magazines.^{[32][33][34]}



From the left, .22 Short, .22 LR, .22 Winchester Magnum Rimfire, and .22 Hornet

Bullets for the .22 WMR are generally unlubricated lead with heavy copper plating, in either solid nose or hollow point style designed for small game hunting or pest control (varmint hunting).

The limited selection of commercial ammunition for the .22 WMR has inspired specialist wildcatters to select the .22 WMR case for handloading high performance rimfire ammunition. Generally they load the wildcat cartridges with pointed bullets for the aerodynamic advantages, using the same bullets as those in .22 caliber centerfire cartridges. Though such bullets are generally heavier than standard .22 WMR, the sharp nose and tapered tail conserve energy better, delivering greater impact at longer ranges.

Other wildcatters neck the .22 WMR down to smaller calibers, such as .20 (5 mm) and .17 (4.5 mm) or even smaller, in an attempt to get maximum velocity and the flattest possible trajectory. An example of such an experimental design is the Swedish 4.5×26mm MKR.

See also

- .22 BB
- .22 CB
- .22 Short
- .22 Long
- .22 Extra Long
- .22 Long Rifle
- .22 Hornet
- 5 mm caliber
- List of handgun cartridges
- List of rifle cartridges
- List of rimfire cartridges

- Table of handgun and rifle cartridges

References

1. *Cartridges of the World 11th Edition*, Book by Frank C. Barnes, Edited by Stan Skinner, Gun Digest Books, 2006, ISBN 978-0-89689-297-2 pp. 490, 492
2. "XT 22 Magnum - 22 Magnum - Armscor International, Inc" (<https://web.archive.org/web/20180813022031/https://armscor.com/firearms/xt-series/xt-22-magnum-22-magnum/>). *armscor.com*. Archived from the original (<http://armscor.com/firearms/xt-series/xt-22-magnum-22-magnum/>) on 13 August 2018. Retrieved 15 April 2018.
3. "Federal Premium Ammunition - Rimfire" (<https://web.archive.org/web/20171205194553/http://federalpremium.com/ammunition/rimfire/family/game-shok/game-shok/757>). *federalpremium.com*. Archived from the original (<http://federalpremium.com/ammunition/rimfire/family/game-shok/game-shok/757>) on 2017-12-05. Retrieved 2017-12-05.
4. "CCI Ammunition - Product Detail" (<http://www.cci-ammunition.com/products/detail.aspx?use=1&loadNo=0059>). *www.cci-ammunition.com*.
5. Barnes 1972, p.275, ".22 Winchester Magnum Rimfire".
6. Frank C. Barnes (2003). Stan Skinner (ed.). *Cartridges of the World* (10th ed.). Krause Publications. ISBN 978-0-87349-605-6.
7. "CCI Ammunition - View All CCI Loads" (<http://www.cci-ammunition.com/products/products.aspx>). *www.cci-ammunition.com*.
8. <https://www.gunsamerica.com/digest/the-m6-springfield-scout-rifle/> Bring Back The M6 Scout Rifle by JON HODOWAY on MARCH 5, 2016
9. <http://www.oldjimbo.com/survival/v-shrake/m6.html> Archived (<https://web.archive.org/web/20210425223834/http://www.oldjimbo.com/survival/v-shrake/m6.html>) 2021-04-25 at the Wayback Machine The M6 Survival System
10. "Chiappa Firearms - M6" (<https://web.archive.org/web/20190311051738/http://chiappafirearms.com/f.php?id=35>). *www.chiappafirearms.com*. Archived from the original (<https://www.chiappafirearms.com/f.php?id=35>) on 2019-03-11. Retrieved 2018-10-28.
11. "Ruger 10/22 Magnum Serial Number History" (<http://www.ruger.com/service/productHistory/RI-10-22M.html>). *Product History*. Sturm, Ruger & Co., Inc. Retrieved 25 March 2013.
12. "Autoloading Rimfire Rifle - Model 597" (<https://web.archive.org/web/20130402104728/http://www.remington.com/en/product-families/firearms/rimfire-families/autoloading-model-597.aspx>). *Remington Guns*. Remington Arms Company, LLC. Archived from the original (<http://www.remington.com/en/product-families/firearms/rimfire-families/autoloading-model-597.aspx>) on 2013-04-02. Retrieved 25 March 2013.
13. Slowik, Max (18 January 2012). "EAA's Latest Appeal: .22 Long Rifle and .22 Magnum Bullpup Carbines" (<https://web.archive.org/web/20140127200734/http://www.guns.com/2012/01/18/eaas-latest-appeal-22-long-rifle-and-22-magnum-bullpup-carbines/>). *Blog*. Guns.com. Archived from the original (<http://www.guns.com/2012/01/18/eaas-latest-appeal-22-long-rifle-and-22-magnum-bullpup-carbines/>) on 27 January 2014. Retrieved 25 March 2013.
14. CTD Mike (8 March 2012). "EAA to Import New .22 Tanfoglio "Appeal" Bullpup Carbine" (<https://archive.today/20130410165335/http://cheaperthandirt.com/blog/?p=19529>). *The Shooters Log*. Cheaper Than Dirt. Archived from the original (<http://cheaperthandirt.com/blog/?p=19529>) on 10 April 2013. Retrieved 25 March 2013.
15. "Taurus Revolver Manual" (https://web.archive.org/web/20130128034340/http://www.taurususa.com/pdf/revolver_manual.pdf) (PDF). *Safety and Operation Manual*. Taurus International. 25 February 2003. Archived from the original (http://www.taurususa.com/pdf/revolver_manual.pdf) (PDF) on 2013-01-28. Retrieved 25 March 2013.

16. B. Gil Horman (24 March 2011). "North American Arms Mini Revolvers" (<http://www.americanrifleman.org/articles/north-american-arms-revolver-review/>). *American Rifleman*. Retrieved 25 March 2013. "The .22 Mag. models can be fitted with a second cylinder chambered in .22 Long Rifle, which makes practice and plinking inexpensive indeed."
17. "Smith & Wesson Air Lite 351PD (.22 Magnum)" (<https://web.archive.org/web/20110308203407/http://www.kansascarry.com/firearms-reviews/4-smith-a-wesson-air-lite-351pd.html>). *Kansas Carry*. Custom Digital Services, LLC. 2010. Archived from the original on 2011-03-08. Retrieved 25 March 2013.
18. NRA Staff (22 April 2011). "Heritage Mfg. Rough Rider Revolvers" (<https://archive.today/20130414014251/http://www.americanrifleman.org/articles/heritage-mfg-rough-rider-revolvers/>). *American Rifleman*. Archived from the original (<http://www.americanrifleman.org/articles/heritage-mfg-rough-rider-revolvers/>) on 2013-04-14. Retrieved 25 March 2013. "Chambered in .22 Long Rifle and .22 WMR, Heritage Mfg.'s nine-shot Rough Rider revolvers are now available in blued, black satin, silver satin and color case-hardened-style finishes."
19. Quinn, Jeff (5 January 2013). "Lightweight, Compact Ruger LCR 22 Magnum Pocket Revolver" (<http://www.gunblast.com/Ruger-LCR22Mag.htm>). *GUNBlast.com*. Retrieved 25 March 2013. "I am glad that Ruger is including the 22 magnum along with the 22 LR, 38 Special Plus P, and 357 magnum chamberings in this handy little revolver."
20. B. Gil Horman (15 February 2011). "The Kel-Tec PMR-30" (<http://www.americanrifleman.org/articles/kel-tec-pmr-30-review/>). *American Rifleman*. Retrieved 25 March 2013. "As a fan of the .22 Mag. cartridge, I am very pleased that Kel-Tec built a lightweight, high-capacity semi-auto that shows off what this neat little cartridge can do."
21. Eger, Chris (7 March 2013). "The Grendel Pistols: The forgotten polymers (and the birth of Kel-tec)" (<https://web.archive.org/web/20130324233533/http://www.guns.com/2013/03/07/grendel-pistols-keltec-kel-tec/>). *Guns.com*. Archived from the original (<http://www.guns.com/2013/03/07/grendel-pistols-keltec-kel-tec/>) on 24 March 2013. Retrieved 25 March 2013. "With the use of polymers, the weight of this target pistol was kept down to 26 ounces when loaded with an impressive array of 30-rounds of rimfire magnums."
22. Kelly, Paco (9 August 2006). "Still Holding On: AMT's AutoMag II .22 Magnum Rimfire Auto" (http://www.gunblast.com/Paco_22AutoMag.htm). *GUNBlast.com*. Retrieved 25 March 2013. "It wasn't very long ago that some very well-designed little .22 Magnum handguns were available from AMT (Arcadia Machine & Tool) of Irwindale, CA."
23. Quinn, Jeff (27 November 2006). "High Standard Brings Back the AMT Automag II .22 Magnum Pistol" (<http://www.gunblast.com/AMT-AutoMagII.htm>). *GUNBlast.com*. Retrieved 25 March 2013. "After being out of production for several years, High Standard Manufacturing is now producing the Automag II pistol that was previously produced by AMT in Irwindale, California and Galena Industries in Sturgis, South Dakota."
24. "AMT AutoMag" (https://web.archive.org/web/20090616032529/http://www.highstandard.com/amt_automag.html). Archived from the original (http://www.highstandard.com/amt_automag.html) on June 16, 2009.
25. Lermayer, Ralph M. (July 2010). "Evolution of the .22" (<http://buckmasters.com/Magazines/GunHunter/Articles/ID/2454/Evolution-of-the-22>). *GunHunter Magazine*. Buckmasters Ltd. Retrieved 17 September 2016. "... take a .22 rimfire smoothbore (like the Mossberg or Marlin's Garden gun), and patterns will be surprisingly good out to 15 yards."
26. "Marlin's .22 Mag Garden Gun" (<http://www.fieldandstream.com/articles/hunting/1999/12/marlin-22-mag-garden-gun>). *Fieldandstream.com*. *Field & Stream*. December 1999. Retrieved 17 September 2016. "A gopher-getter? Get serious."
27. Rees, Clair (March 2000). "Marlin's 'Garden Gun' - Model 25MG" (https://web.archive.org/web/20151107062919/http://www.findarticles.com/p/articles/mi_m0BQY/is_3_46/ai_59281208). *Guns Magazine*. Archived from the original (http://findarticles.com/p/articles/mi_m0BQY/is_3_46/ai_59281208/) on November 7, 2015. Retrieved 17 September 2016. Alt URL (<http://www.thefreelibrary.com/Marlin's+%22Garden+Gun%22.-a059281208>)

28. Eger, Christopher (28 July 2013). "Marlin 25MG Garden Gun" (<https://web.archive.org/web/20160918005923/http://www.marlinforum.com/Marlin-25MG-Garden-Gun.html>). *Marlin Firearms Forum*. Outdoor Hub LLC. Archived from the original (<http://www.marlinforum.com/Marlin-25MG-Garden-Gun.html>) on 2016-09-18. Retrieved 17 September 2016.
29. Guncraft 22 Magnum AR15 Review (<https://www.gunmart.net/gun-reviews/firearms/rifles/guncraft-22-magnum-ar15>) Pete Moore. GunMart. October 14, 2019.
30. Barnes 1972, p.275, ".22 Winchester Rimfire (WRF)"
31. Sparano, Vin (2000). *The Complete Outdoors Encyclopedia*. New York: St. Martin's Press. ISBN 978-0-312-26722-3.
32. "Remington Magnum Rimfire: 22 Win Mag PR22M1 33-grain AccuTip-V" (https://web.archive.org/web/20080530170242/http://www.remington.com/products/ammunition/rimfire/magnum_rimfire.asp). Remington Arms Company, Inc. Archived from the original (http://www.remington.com/products/ammunition/rimfire/magnum_rimfire.asp) on 2008-05-30. Retrieved 2008-06-17.
33. "CCI: Varmint 0073 22 Magnum RF V-MAX 30-grain poly-tip" (<https://web.archive.org/web/20080516032801/http://www.cci-ammunition.com/ballistics/rimfire.aspx>). CCI. Archived from the original (<http://www.cci-ammunition.com/ballistics/rimfire.aspx>) on 16 May 2008. Retrieved 2008-06-17.
34. "Hornady: 22 WMR 30-grain V-MAX" (https://web.archive.org/web/20090210190834/http://www.hornady.com/shop/?page=shop%2Fbrowse&category_id=9b26ee2729baf268e4ae1c61acde17cc). Hornady Manufacturing, Inc. Archived from the original (http://www.hornady.com/shop/?page=shop%2Fbrowse&category_id=9b26ee2729baf268e4ae1c61acde17cc) on 2009-02-10. Retrieved 2008-06-17.

Further reading

- Barnes, Frank C., ed. by John T. Amber. ".22 Winchester Magnum Rimfire", in *Cartridges of the World*, pp. 275. Northfield, IL: DBI Books, 1972. ISBN 978-0-695-80326-1.
- *Cartridges of the World 11th Edition*, Book by Frank C. Barnes, Edited by Stan Skinner, Gun Digest Books, 2006, ISBN 978-0-89689-297-2 p. 479

External links

- Chuck Hawks' page (<http://www.chuckhawks.com/22mag.htm>) on the .22 WMR
- Chuck Hawks (http://www.chuckhawks.com/ammo_roundup_22WMR.htm) on .22 WMR ammunition selection
- The .22 WMR TODAY (http://www.findarticles.com/p/articles/mi_m0BQY/is_6_47/ai_74033120) Archived (https://web.archive.org/web/20151107113851/http://www.findarticles.com/p/articles/mi_m0BQY/is_6_47/ai_74033120) 2015-11-07 at the Wayback Machine, Guns Magazine, June, 2001 by C. Rodney James
- Three .22 WMR Loads From Remington (https://web.archive.org/web/20120314060826/http://www.handgunsmag.com/2010/11/11/ammunition_wmr_072804/), Handguns Magazine, by Phil W. Johnston
- Reloading Techniques: .22 Win. Magnum Rimfire (<https://web.archive.org/web/20070403101444/http://www.gun-tests.com/performance/apr96reloading.html>), Performance Shooter, April 1996

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